4.02 Complex Numbers				
4.02a	The language of complex numbers	a) Understand the language of complex numbers. *Know the meaning of "real part", "imaginary part", "conjugate", "modulus" and "argument" of a complex number.*		B2 B5 B3
4.02b 4.02d		b) Be able to express a complex number z in either cartesian form $z = x + \mathrm{i}y$, where $\mathrm{i}^2 = -1$, or modulus-argument form $z = r(\cos\theta + \mathrm{i}\sin\theta) = [r,\theta] = r\mathrm{cis}\theta$, where $r \geq 0$ is the modulus of z and θ, measured in radians is the argument of z.	d) Understand and be able to use the exponential form, $r\mathrm{e}^{\mathrm{i}\theta}$, of a complex number.	B9
4.02c		c) Understand and be able to use the notation: $z, z^*, \mathrm{Re}(z), \mathrm{Im}(z), \arg(z), \lvert z \rvert$. *Includes knowing that a complex number is zero if and only if both the real and imaginary parts are zero.* *The principal argument of a complex number, for uniqueness, will be taken to lie in either of the intervals* $[0, 2\pi)$ *or* $(-\pi, \pi]$. *Learners may use either as appropriate unless the interval is specified.* *In stage 1 knowledge of radians is assumed: see H240 section 1.05d.*		

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.02e 4.02f	Basic operations	e) Be able to carry out basic arithmetic operations $(+,-,\times,\div)$ on complex numbers in both cartesian and modulus-argument forms. *In stage 1 knowledge of radians and compound angle formulae is assumed: see H240 sections 1.05d and 1.05l.* *Learners may use the results* $z_1z_2 = [r_1r_2, \theta_1 + \theta_2]$ *and* $\frac{z_1}{z_2} = \left[\frac{r_1}{r_2}, \theta_1 - \theta_2\right]$. f) Convert between cartesian and modulus-argument forms.		B2 B5 B6
4.02g 4.02h 4.02i 4.02j	Solution of equations	g) Know that, for a polynomial equation with real coefficients, complex roots occur in conjugate pairs. h) Be able to find algebraically the two square roots of a complex number. *e.g. By squaring and comparing real and imaginary parts.* i) Be able to solve quadratic equations with real coefficients and complex roots. j) Be able to use conjugate pairs, and the factor theorem, to solve or factorise cubic or quartic equations with real coefficients. *Where necessary, sufficient information will be given to deduce at least one root for cubics or at least one complex root or quadratic factor for quartics.*		B1 B3

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.02k 4.02m 4.02l	Argand diagrams	k) Be able to use and interpret Argand diagrams. *e.g. To represent and interpret complex numbers geometrically.* *Understand and use the terms "real axis" and "imaginary axis".* l) Understand the geometrical effects of taking the conjugate of a complex number, and adding and subtracting two complex numbers.	m) Understand the geometrical effects of multiplying and dividing two complex numbers. *Includes raising complex numbers to positive integer powers.*	B4 B6
4.02n	Euler's formula		n) Know and be able to use Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$. *e.g. To express, and work with, complex numbers in the forms* $r(\cos\theta + i\sin\theta) = re^{i\theta} = r\text{cis}\,\theta = [r,\theta]$.	B6 B9

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.02o	Loci	o) Be able to illustrate equations and inequalities involving complex numbers by means of loci in an Argand diagram. *i.e. Circle of the form* $\lvert z-a\rvert = k$, *half-lines of the form* $\arg(z-a)=b$, *lines of the form* $\mathrm{Re}(z)=k$, $\mathrm{Im}(z)=k$ *and* $\lvert z-a\rvert = \lvert z-b\rvert$, *and regions defined by inequalities in these forms.* *To include the convention of dashed and solid lines to show exclusion and inclusion respectively.* *No shading convention will be assumed. If not directed, learners should indicate clearly which regions are included.* *In stage 1 knowledge of radians is assumed: see H240 section 1.05d.*		B7
4.02p		p) Understand and be able to use set notation in the context of loci. *e.g. The region* $\lvert z-a\rvert > k$ *where* $z = x + \mathrm{i}y$, $a = x_a + \mathrm{i}y_a$ *and* $k > 0$ *may be represented by the set* $\{x+\mathrm{i}y : (x-x_a)^2 + (y-y_a)^2 > k^2\}$. *In stage 1 knowledge of radians is assumed: see H240 section 1.05d.*		

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
4.02q	De Moivre's theorem		q) Understand de Moivre's theorem and use it to find multiple-angle formulae and sums of series involvin trigonometric and/or exponential terms. *Express trigonometrical ratios of multiple angles in terms c powers of trigonometrical ratios of the fundamental angle* *e.g.* $\sin(3\theta) = 3\sin\theta - 4\sin^3\theta$. *Use expressions for* $\sin\theta$ *and* $\cos\theta$ *in terms of* $e^{i\theta}$ *or equivalent relationships.* *e.g.* $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$. *Express powers of* $\sin\theta$ *and* $\cos\theta$ *in terms of series of trigonometric ratios of multiples of the fundamental angle* *e.g.* $\sin^5\theta = \frac{1}{16}(10\sin\theta - 5\sin(3\theta) + \sin(5\theta))$.
4.02r	nth roots		r) Be able to find the n distinct nth roots of $re^{i\theta}$ for $r \neq$ and know that they form the vertices of a regular n-gon on an Argand diagram. *Answers may be asked for in either cartesian or modulus-argument form.*
4.02s	Roots of unity		s) Be able to use complex roots of unity to solve geometric problems. *e.g. To locate the roots of unity on an Argand diagram or t prove results about sums of roots of unity.*